

In Search of Search

Soon, you're going to have mountains of information—both on your hard disk and on the Net—and a terrible time finding anything. Or will you?

The world produces up to 2 exabytes of data every year—that's 2,00,00,00,000 gigabytes. Martin Stone at ComputerUser.com predicts, "It's taken 3,00,000 years for humans to accumulate 12 exabytes of information. It will take just 2.5 more to create the next 12 exabytes." How do we make use and sense of all this data? How do we archive and track it? How much of it is useful? These are pressing questions, now more than ever before.

All this data resides on personal hard disks, corporate databases, CD-ROMs and of course, on the Web. Most of it is unstructured—it is in free-text form, unlike structured data, which is readily machine-readable—a filled-in electronic form, for example. The unstructured nature of most data means that retrieving information is a non-trivial business. There is more data out there than we can easily make sense of, which is why there is a lot being done along this direction.

Metadata and search

Metadata is data about data; it can be as simple as just the author, date and time a document was created, or be as informative as a list of keywords and key-phrases that describe what the document is about. Sufficient metadata would remove the ambiguity that typically occurs in a search. For example, a search on 'personal search' could mean both a search for a dating service, as well as a search for a personalised search engine.

One of the reasons that search engines are not as efficient as they could be, is that not enough people bother tagging their Web pages with metadata. If every author of every Web page annotated the page with metadata that documented exactly what the page was all about, search might have been a rather simple business.

In its early days, Yahoo! used metadata to manually index pages; Google uses the PageRank algorithm, which seems to work pretty well. Also, Google computes metadata automatically,

not manually. Others, like Vivisimo, generate metadata automatically, and auto-categorise pages by generating a taxonomy.

The Web currently lacks standards for metadata, which would make the task of automated indexing easier. The most advanced initiative to address this problem is the Dublin Core Metadata Initiative, which is geared towards attaching metadata to files for indexing systems to collect. A set of metadata elements have been created by the programme, as well as methods for incorporating them in Web pages. The metadata can be derived by humans as well as crawlers. In a similar vein, the Yahoo database is a commercial project that classifies Web sites by subject area. The Dublin Core has become an important part of the emerging infrastructure of the Internet.

Semantics in search

The problems with search engines, as we've all experienced, are all too